

KKT Conditions as Dual Certificate (Sec. 13.6)

Condition	Requirement	Interpretation
Primal Feasibility	$\mathbf{C}_{LE}\mathbf{x} \leq \mathbf{d}_{LE}, \mathbf{C}_{EQ}\mathbf{x} = \mathbf{d}_{EQ}$	$\mathbf{x}$ is in the feasible set
Dual Feasibility	$\boldsymbol{\mu} \geq \mathbf{0}$	Dual variables are non-neg.
Complementary Slackness	$\boldsymbol{\mu}^\top (\mathbf{C}_{LE}\mathbf{x} - \mathbf{d}_{LE}) = 0$	Active iff constraint is tight
Zero Duality Gap	$f(\mathbf{x}^*) = D(\boldsymbol{\mu}^*, \boldsymbol{\lambda}^*)$	Confirms global optimality